FinAgentBench: Separating Execution from Judgment in LLM Financial Research Agents

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Abstract

Can large language models do quantitative finance research? We argue this question conflates two distinct capabilities and measure them separately across up to 23 frontier models run through one identical agent harness. *Execution* — replicating a quantitative result from a written method specification and raw data — is essentially solved: on four replication tasks of increasing difficulty, nearly every model produces exact results, with a 200× cost spread between the cheapest and most expensive perfect replication; only a fifth task at the scale of an academic machine-learning paper splits the field (9 of 22 models perfect). *Judgment* — predicting returns from financial text — requires care to measure: on pre-training-cutoff data, models achieve rank correlations up to 0.53 with future stock returns *with the text withheld*, revealing pervasive memorization of market history that contaminates naive backtests. On text published strictly after training cutoffs, genuine judgment emerges with a steep capability ladder (information coefficients of 0.13–0.59) and a sharp generational jump in the newest frontier models. Signal exists only where information diffuses slowly: analysts' opinion essays and institutional holdings disclosures predict returns; earnings releases, call transcripts, and central-bank statements — instantly priced — do not. Simulated long-short portfolios on the strongest signal are positive in all ten configurations tested, and we pre-register live predictions to enable out-of-sample verification. The complete study consumed roughly $730 of inference credits and 170 million tokens, a cost structure that changes who gets to run capability experiments.

1 Introduction

Large language models increasingly operate as autonomous agents inside financial-research workflows. Yet claims about what they can do tend to blur two very different capabilities. The first is

*Agentic Sciences. Contact: qr33@cornell.edu. Live results, plain-language report, and pre-registered predictions: `https://qihongruan.github.io/finagentbench/`. All experiments were orchestrated end-to-end by an autonomous coding agent over three days on a single inference platform; every reported number is generated programmatically from run logs.

execution: given a research recipe — filters, windows, formulas — and raw data, write the code and produce correct numbers. This is the work of a research assistant. The second is *judgment*: read a piece of financial text and correctly anticipate what the market does next. This is the work of an analyst, and it is rare in humans as well.

Survey evidence indicates that practitioners deploying production agents lack objective evaluations: 74% rely primarily on human evaluation, and reliability is their top development challenge (Pan et al., 2026). This paper contributes a fully programmatic benchmark that measures both capabilities separately, with three design principles.

Private ground truth. All replication targets are computed on private data extracts (custom sample windows of CRSP, a consolidated options tape, and proprietary panels), so published numbers cannot be recalled from training corpora; a scripted grader compares each submitted statistic to ground truth within tolerance bands. No LLM judges are used anywhere.

Withheld-text controls. Every text-prediction task is run twice per model: once with the document and once with it withheld (the model sees only ticker and date). The difference isolates what the model extracts from *reading* as opposed to *remembering*. We show this control is not optional: without it, memorization masquerades as skill.

Post-cutoff samples. Our central judgment test uses documents published April–August 2026, after the training cutoffs of the models under test (verified per-model from official documentation where disclosed), plus a within-sample clean-window comparison as a contamination diagnostic.

Our findings, previewed in the abstract, connect to several literatures: momentum and its replication (Jegadeesh and Titman, 1993), machine-learning asset pricing (Gu et al., 2020), post-publication decay of predictors (McLean and Pontiff, 2016), and textual analysis in finance (Tetlock, 2007; Loughran and McDonald, 2011). Relative to the emerging literature evaluating LLMs on financial prediction, our contribution is methodological as much as empirical: we demonstrate that the standard design — backtest an LLM on historical text — is confounded by memorized outcomes, and we provide two remedies (controls and post-cutoff sampling) cheap enough for any study to adopt.

2 Benchmark design

2.1 Harness

Every model runs through one identical harness against an OpenAI-compatible inference platform hosting all models under a single account. For replication (Track A), the agent receives a task sheet (method fully specified, results withheld), read-only data paths, a bash sandbox with a scientific Python stack, and a step budget (40 steps; 60 for the hardest task). Two tools are exposed: `run_bash` and `submit_answer`. Grading is scripted: a submitted statistic passes "loose" within 5% relative error and "strict" within 1% (wider bands for stochastic neural-network quantities). For text prediction (Track B), each document is a single prediction call with a JSON response schema; scoring uses hit rates, Spearman rank information coefficients (IC), and long-short decile or tercile spreads.

Running 23 heterogeneous models through one harness surfaced seven provider-specific API incompatibilities (parameter names, reasoning-token echo requirements, signed-message constraints, backend-pool routing lotteries); we document them in the repository because they account for a material share of observed "agent unreliability" — consistent with practitioners' reports that reliability problems are substantially systems-level (Pan et al., 2026).

2.2 Track A: five replication tasks

- **T1** Cross-sectional momentum on a private 20-stock mega-cap panel, 2004–2024 (ground truth Sharpe 0.386).
- **T2** A news-event study with filtering, deduplication, winsorization, and a Welch test (spread 23.6bps, $t = 4.37$).
- **T3** A covered-call backtest on option close panels with mark-to-market rules, missing-quote filling, and sample-gap chaining.
- **T4** Full-market 12–2 momentum deciles computed from an 11GB daily CRSP file (long-short 8.0%/yr, Sharpe 0.43, 2005–2024).
- **T6** A scoped replication of Gu et al. (2020): 94 firm characteristics, rank-normalized monthly; OLS-3, elastic net, gradient-boosted trees, and a three-layer neural network; expanding-window annual refits; out-of-sample 2015–2021. The agent must train all four models and report out-of-sample R^2 and decile long-short Sharpe ratios.

2.3 Track B: seven text tracks

Each track pairs documents with market-adjusted forward returns and includes a withheld-text control. **B1**: 1,008 stratified news headlines (2004–2024) from a commercial institutional news-event dataset, 5-day beta-adjusted returns. **B2**: 297 full-text investment-research essays from a large investment-research platform, 284 published April–August 2026 (post-cutoff), 20-day returns. **B3**: 204 MD&A sections from 10-K/20-F filings (2017–2024, SEC EDGAR), 20-day post-filing drift. **B4**: 100 earnings press releases (8-K item 2.02, Feb–Jul 2026, EDGAR), 5-day returns. **B5**: 38 top position increases from Q1-2026 13F filings of 11 prominent funds, 20-day returns. **B6**: 99 earnings-call transcripts (Feb–Jul 2026), 5-day returns. **B7**: 42 FOMC statements and minutes (2024–2026, Federal Reserve), 5-day SPY and TLT reactions.

Table 1: Memorization on pre-cutoff data: Spearman IC with the document shown versus withheld. Selected models; full tables online.

	IC		
Track / model	Text shown	Text withheld	Δ
B1 headlines (n=1,008), strongest model	0.522	0.533	−0.011
B1, median of 13 models	0.137	0.044	+0.03
B7 FOMC (n=42), strongest model	0.708	0.713	−0.005
B3 MD&A (n=204), strongest model	0.547	0.443	+0.104

3 Results

3.1 Execution is solved — until academic-paper scale

On T1–T4, models are near-uniformly correct. T1: all 13 initially tested models scored strict-perfect, later extended to 21 of 23 (two failures were platform API errors, not capability); the cheapest perfect replication cost $0.001. T2: 13/13. T3, the option backtest with eight interacting specification rules: 12/13 (the single miss submitted after two steps with several fields wrong). T4, requiring database-scale engineering: 10/13 strict. Differentiation appears only at T6, the machine-learning-paper replication: 9 of 22 models reproduced every statistic within tolerance; the most common failure mode was correct model training with subtly wrong portfolio accounting. Notably, success at T6 is not predicted by price tier: several mid-priced models succeeded where flagship models failed.

Reliability re-runs (five seeds on T1–T2, three on T6) found genuine “confidently wrong” episodes in roughly 2.5% of submitted runs, concentrated in a few models, plus a 6% infrastructure failure rate. In production terms both count against dependability, and both are invisible in single-run leaderboards.

3.2 Models have memorized market history

Table 1 shows the memorization result. On B1 (historical headlines), the strongest model achieves IC 0.52 reading headlines — and IC 0.53 *with headlines withheld*, ranking 20 years of stock outcomes from ticker and date alone (67% directional accuracy). On B7, the same model “predicts” the market’s five-day reaction to past FOMC releases at 90% directional accuracy with the release text hidden. On B3 (historical MD&A), its with-text IC of 0.55 decomposes into 0.44 of memory and +0.10 of genuine reading increment. Across models, the text-minus-control increment on historical data ranges from −0.03 to +0.09: for most models, historical “text prediction” is dominated by recall.

The implication is methodological: any evaluation of LLM return prediction on data preceding the model’s training cutoff is uninterpretable without a withheld-text control, and most published demonstrations lack one.

Table 2: Judgment by text type. IC ranges are across models (main runs); "memory" indicates the withheld-text control is materially positive.

Track	Sample	IC range	Verdict
B2 analyst essays	297 (post-cutoff)	0.13 – 0.59	Real signal
B5 13F position adds	38 (post-cutoff)	−0.16 – 0.58	Positive, small n
B4 earnings releases	100 (post-cutoff)	−0.19 – 0.08	None
B6 call transcripts	99 (post-cutoff)	−0.16 – 0.00	None
B7 FOMC releases	42 (mostly pre-cutoff)	—	Memory
B1 news headlines	1,008 (pre-cutoff)	—	Mostly memory
B3 MD&A sections	204 (pre-cutoff)	—	Mostly memory

3.3 On post-cutoff text, judgment is real, laddered, and jumping

B2 is the clean test: essays published after the models' training cutoffs (verified per model; five models with undisclosed cutoffs are flagged, and a June–July-only sub-window provides a contamination diagnostic — for no model does the clean window show materially weaker performance, the signature contamination would produce). Withholding the essay collapses every model to noise; predictions are stable across repeated sampling (rank correlation ≈ 0.9 between independent runs, ICs within 0.05).

The ladder is steep: small open-weight models score IC 0.13–0.19; mid-tier frontier models 0.21–0.33; the strongest current models 0.39–0.59 (n=297; the top score's standard error is roughly 0.05). Within one vendor family we observe four successive versions plateaued at IC 0.26–0.28 before the newest version doubled to 0.59; a sibling product line instead climbed gradually ($0.22 \rightarrow 0.32 \rightarrow 0.39$). Capability in this task did not accumulate smoothly — it arrived in a jump.

A caveat on calibration: several models combine strongly positive rank ICs with directional accuracy *below* 50% in a window where only 39% of essays preceded outperformance — models rank well but retain a long bias. Practical use is long-short, not directional.

3.4 Which text carries alpha

Table 2 summarizes the seven tracks. The pattern is economically coherent: predictability lives where information diffuses slowly — interpretive essays and delayed institutional disclosures — and is absent where markets react within minutes of the event (earnings releases, call transcripts, FOMC announcements), consistent with prices rapidly impounding hard public information and with slower diffusion of soft information (Tetlock, 2007; Loughran and McDonald, 2011; Ball and Brown, 1968). On earnings releases, most models' naive "good quarter → buy" mapping produces mildly *negative* next-day-entry ICs.

3.5 From ranking to portfolios, and a pre-registered test

Converting B2 predictions into weekly long-short tercile portfolios (20-day overlapping holds, 10bps one-way costs), all ten configurations tested — eight single models and two ensembles — were positive over the 67 trading days available, with annualized Sharpe ratios between 0.2 and 3.6. We emphasize the sample is three months, concentrated in technology names, with overlapping cohorts; we treat this as consistency evidence, not an alpha estimate. To enable genuine out-of-sample evaluation, on August 13, 2026 we locked predictions from five models for 43 just-published essays into the public repository history before any outcomes existed; results will be scored and published, favorable or not, in September 2026.

3.6 A side finding: machine-learning alpha after publication

Constructing T6's ground truth required re-running a scoped Gu et al. (2020) pipeline on 2015–2021 — entirely after the paper's out-of-sample period. Equal-weighted decile long-short Sharpe ratios fall from above 2 in the published era to 0.20 (boosted trees) and 0.32 (neural network), with linear models turning negative (-0.10, -0.31); out-of-sample R^2 remains small and positive for linear and tree models. The qualitative ranking — nonlinear beats linear — survives; the economic magnitude largely does not, consistent with post-publication decay (McLean and Pontiff, 2016). (Scope deviations from the original: characteristics only, equal weights, training histories beginning 2004.)

4 Cost

The complete study — five replication tasks across up to 23 models with multi-seed reliability re-runs, seven text tracks totaling roughly 1,800 labeled events with controls and repeated seeds, portfolio simulations, and a pre-registered live round — consumed approximately $730 of inference credits: 170 million tokens across 63,000 prediction calls and about 400 agent episodes, orchestrated autonomously over three days. Agent labor is nearly free (the entire Track A grading pipeline cost about $40); judgment elicitation at scale is what costs money. For comparison, a contemporary human-methods study of production agents required 25 authors, a 306-respondent survey, and roughly a year (Pan et al., 2026) — a fully-loaded cost we estimate at low six figures. Both designs answer questions the other cannot; but for questions of measurable capability, the marginal cost of simply running the experiment has fallen by roughly three orders of magnitude.

5 Limitations

(i) B2's post-cutoff guarantee is per-model and depends on disclosed cutoffs; five models do not disclose them, and stated cutoffs bound pre-training, not post-training, exposure. Our clean-window diagnostic and controls bound but do not eliminate residual risk. (ii) Track B forward windows overlap in calendar time and cluster in technology names; naive standard errors overstate precision, and we interpret cross-model *rankings* on a common sample rather than absolute significance. (iii) The portfolio simulation spans three months in one regime. (iv) Some tracks are small (B5: n=38; B7: n=42). (v) Single-platform inference may differ from first-party APIs in quantization and serving configuration. (vi) Text datasets are licensed; we publish aggregates only, which limits third-party re-scoring to the pre-registered round.

6 Conclusion

Execution and judgment are different capabilities on different trajectories. The research-assistant layer of quantitative finance is effectively commoditized — perfect replications for pennies, bounded mainly by reliability engineering. Judgment exists, is measurable only with memorization-aware designs, concentrates in the newest frontier models, and depends more on *what* is read than on *which* model reads it. The methodological lesson generalizes beyond finance: for any domain whose outcomes are in the training corpus, capability evaluation requires withheld-input controls and post-cutoff samples. Both are cheap. We provide both, plus a falsifiable forward test.

References

Ball, R., and P. Brown. 1968. An empirical evaluation of accounting income numbers. *Journal of Accounting Research* 6:159–178.

Gu, S., B. Kelly, and D. Xiu. 2020. Empirical asset pricing via machine learning. *Review of Financial Studies* 33:2223–2273.

Jegadeesh, N., and S. Titman. 1993. Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance* 48:65–91.

Loughran, T., and B. McDonald. 2011. When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *Journal of Finance* 66:35–65.

McLean, R. D., and J. Pontiff. 2016. Does academic research destroy stock return predictability? *Journal of Finance* 71:5–32.

Pan, M. Z., N. Arabzadeh, R. Cogo, et al. 2026. Measuring agents in production. arXiv:2512.04123. ICML 2026 (oral).

Tetlock, P. C. 2007. Giving content to investor sentiment: The role of media in the stock market. *Journal of Finance* 62:1139–1168.